

Mode1 - Handshaking

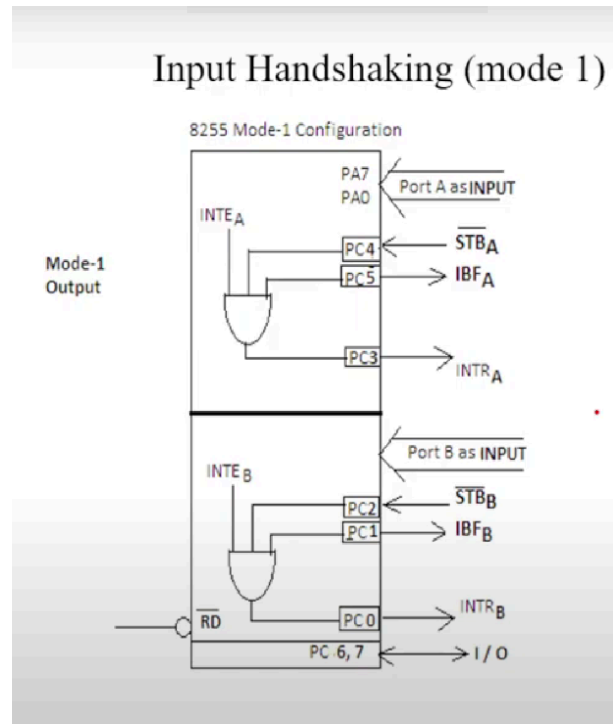
Input Handshaking:

Steps (PORT A):

1. Device sends 8 bit data to data buffer of the IC through PA0-PA7
2. Immediately the strobe signal ($\overline{STB_A} = 0$) is sent from device to IC through PC4 to let the IC know that data is being sent
3. Once the data bus buffer is filled with data the input buffer full ($IBF_A = 1$) signal is sent from IC to device through PC5
4. $\overline{STB_A} = 1$
5. $INTR_A = 1$ is sent from PC3 from IC to MPU for receiving the data
6. If MPU wants, $\overline{RD} = 0$ and reads the data
7. $INTR_A = 0$, $IBF_A = 0$

Steps (PORT B):

1. Device sends 8 bit data to data buffer of the IC through PB0-PB7
2. Immediately the strobe signal ($\overline{STB_B} = 0$) is sent from device to IC through PC2 to let the IC know that data is being sent
3. Once the data bus buffer is filled with data the input buffer full ($IBF_B = 1$) signal is sent from IC to device

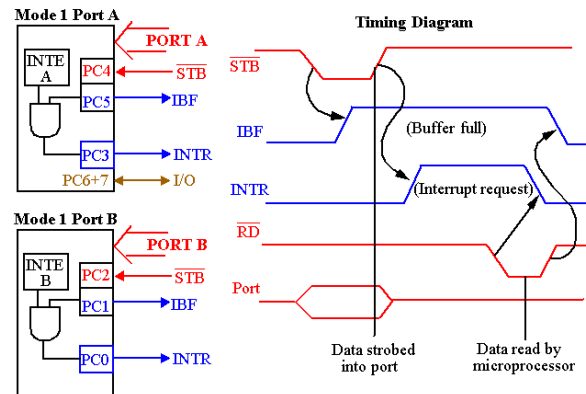


through PC1

4. STB B = 1
5. INTR B = 1 is sent from PC0 from IC to MPU for receiving the data
6. If MPU wants, RD = 0 and reads the data
7. INTR B = 0, IBF B = 0

Timing diagram of input handshaking (Mode 1):

<u>STB</u>	The strobe input loads data into the port latch on a 0-to-1 transition
<u>IBF</u>	Input buffer full is an output indicating that the input latch contain information
<u>INTR</u>	Interrupt request is an output that requests an interrupt
<u>INTE</u>	The interrupt enable signal is neither an input nor an output; it is an internal bit programmed via the PC4(port A) or PC2(port B) bits.
<u>PC7,PC6</u>	The port C pins 7 and 6 are general-purpose I/O pins that are available for any purpose.



Output Handshaking:

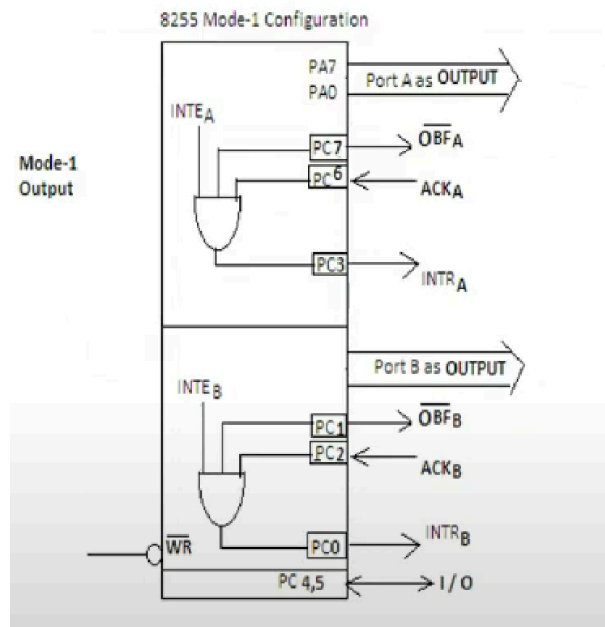
Steps (PORT A):

1. $\text{INTR_A} = 1$ is sent from PC3 from the device via IC to the MPU to check if there is any data to output.
2. If there is, MUP -> $\text{WR} = 0$ and sends data to the data bus buffer of the IC
3. IC sends the MPU output buffer full signal ($\text{OBF_A} = 0$) through PC7 to tell the device to take data
4. $\text{INTR_A} = 0$
5. Once the device is ready it sends $\text{ACK_A} = 0$ to the IC through PC6
6. The device then takes data from the data bus buffer through PA0-PA7 for providing output
7. Once all data is taken, $\text{OBF_A} = 1$, $\text{INTR_A} = 1$

Steps (PORT B):

1. $\text{INTR_B} = 1$ is sent from PC0 from the device via IC to the MPU to check if there is any data to output
2. If there is, MUP -> $\text{WR} = 0$ and sends data to the data bus buffer of the IC
3. IC sends the MPU output buffer full signal ($\text{OBF_B} = 0$) through PC1 to tell the device to take data

Output Handshaking (mode 1)



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ACK will be an active low pin in the image

4. INTR_B = 0 5. Once the device is ready it sends <u>ACK_B</u> = 0 to the IC through PC2 6. The device then takes data from the data bus buffer through PB0-PB7 for providing output 7. Once all data is taken, <u>OBF_B</u> = 1, INTR_B = 1	

In mode 2, only bi-directional I/O devices can be connected. In this case the MPU hands over the control to the 82C55 IC right after only sending the control bits to the IC. With each clock pulse the IC configures itself either in input mode or output mode. When the device inputs any data the IC processes it during the input clock pulse and when the device needs to output any data the IC processes it during output clock pulse.

Input & Output handshaking:

Case 1: when port A is connected to a High-level bidirectional device and port B and port C is connected to low level device

Control Bits:

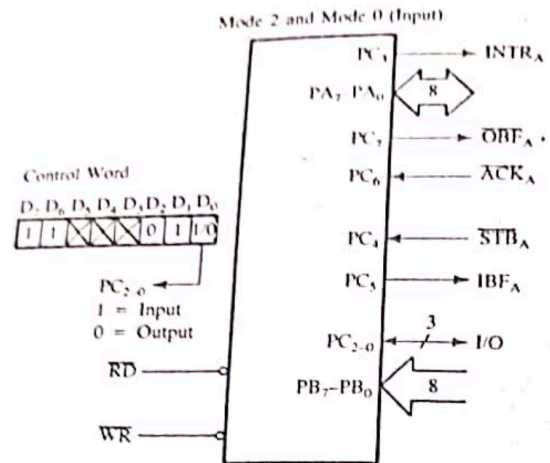
1	1	x	x	x	0	1/0	1/0
D7	D6	D5	D4	D3	D2	D1	D0

D1 - 1 - for input device
0 - for output device

D0 - 1 - for input device
0 - for output device

D3 - Fully dedicated to port A for handshaking

Steps: The Signals work just as mode 1 during the Input/ Output clock pulse of the 82C55 IC with different pin configuration.
Explain by yourself



Case 2: when port A is connected to a High-level bidirectional device and port B is connected to high-level unidirectional device

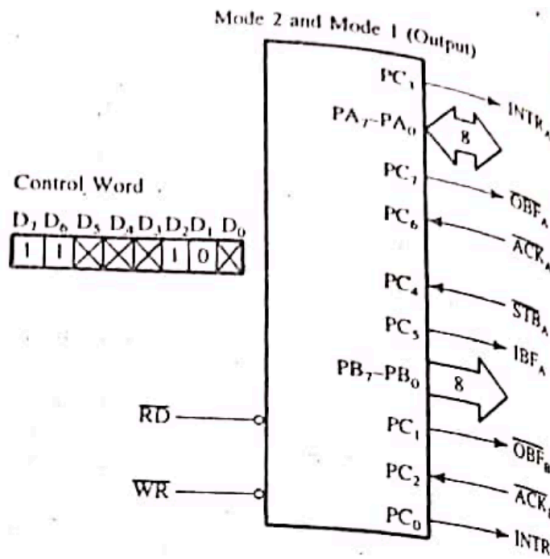
Control Bits:

1	1	x	x	x	1	1/0	x
D7	D6	D5	D4	D3	D2	D1	D0

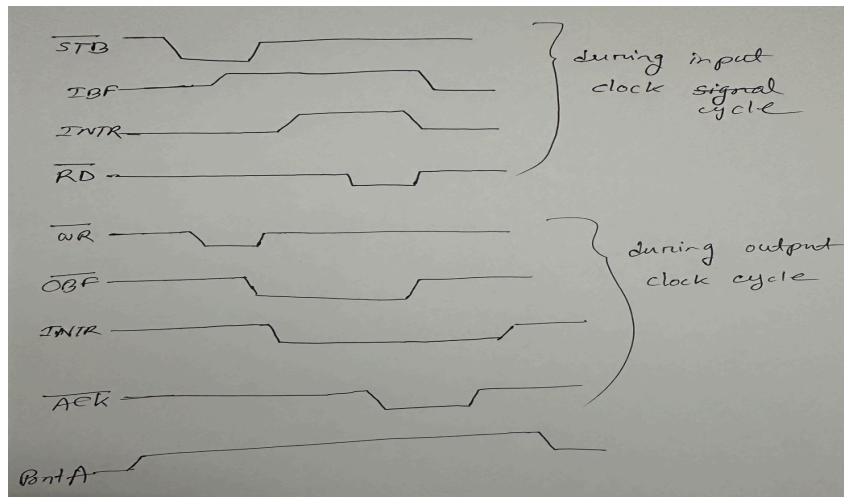
D1 - 1 - for input device
0 - for output device

D3 & D0 - Fully dedicated to port A and Port B for handshaking

Steps: The Signals work just as mode 1 during the Input/ Output clock pulse of the 82C55 IC with different pin configuration.
Explain by yourself



Timing diagram of handshaking (Mode 2):



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